

ATI, too, has announced its DX 10 solution, codenamed the R600. The card is christened X2900XTX/XT and features 512 MB of GDDR3 memory—512 bits wide. The core itself comprises an unbelievable 320 Stream Processors as opposed to 128 on the 8800GTX. Unbelievably, the core is clocked at around 740 MHz!

The term “Stream Processor” is simply the new-fangled name for what has been known as the “Shader Unit.” The X2900XT was released very recently, and while initial performance reviews place it behind the mighty G80, we expect to see lots of performance hikes with newer driver revisions. It’s also possible that DX10 titles may see more efficient use of those extra stream processors, which they were built for anyway...

As always we’re not commenting on its capabilities till we test it ourselves. Details for mid-range and low-end DX10 from ATI are also out.

The last card released from either manufacturer before the X2900XT is the 8800 Ultra from NVIDIA which is clocked higher than the 8800GTX and faster by a hairsbreadth. After the X2900 series release, we could very well see an 8900GTX/GTS, which will probably feature GDDR4. Then there’s a strong rumour that NVIDIA will succeed the dual-PCB, dual-GPU 7950GX2 with an 8950GX2, built around two 8800GTX cores. ATI hasn’t gone dual-GPU, but we’ve seen Sapphire (an ATI-only vendor) unveil one such card—the X1950 Pro Dual, based on two 80nm Radeon X1950 Pro GPU cores.

NVIDIA’s 8-series cards have raised the bar in games with a lot more visual quality than earlier-generation cards, courtesy the immense shader horsepower and bandwidth at their disposal. Improved anti-aliasing and Anisotropic Filtering algorithms bring the latest games to life, and even breathe a touch of life into earlier-generation hits like *Far Cry*, *Doom 3*, and *F.E.A.R.*

GPUs are set to do a lot more than just accelerate games: NVIDIA is showcasing what they call CUDA technology which enables their graphics cards to be used as stream processing engines and floating point operators. The GPU has just invaded the CPU space...

For multimedia buffs, we have HDMI (High Definition Multimedia Interface) support being incorporated in the latest graphics cards. These sport improved hardware acceleration for a better video viewing experience. HDCP (High-definition Digital Content Protection) support is also something to look out for on the latest cards.

HD content (especially 1080p) on integrated solutions is an exercise in futility: a video card is mandatory. Yet another catalyst for the discrete graphics revolution is Vista, whose acclaimed Aero interface has onboard video solutions quivering in their boots. Then there are upcoming game titles like *Crysis*, *BioShock*, and *Time Shift* to name a few, that promise to take terms like realism and immersion to all-new levels.

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Chipsets

You can’t talk about processors without talking about chipsets in the next breath. Here’s all that talk about sockets, pins and DDR3

While talking about the fastest processors, high bandwidth, low-latency memory and the newest batch of Direct X10 cards, we often forget the component that brings all these products together under a single roof. The motherboard is very little more than the sum of its chipsets (the Northbridge and Southbridge) with a hunk of silicon (PCB) thrown in to hold everything. It’s this motherboard that dictates compatibility by allowing or restricting use of certain processors, memory, and other components.

It may seem therefore that the motherboard dictates the choice of the other components. Wrong!

Motherboard chipsets have been evolving, leading to improved products, supporting newer processors primarily, as well as other allied technologies. However, make no mistake—it’s been the processor industry that has largely dictated terms to chipset vendors, even memory vendors for that matter.

Flashback: Intel moved from a 478-pin processor socket to an LGA 775 processor socket.

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